Tangent-Space Consistency: Unsupervised Hallucination Detection via Local Manifold Geometry in Large Language Models

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Abstract

Large Language Models (LLMs) frequently exhibit *hallucinations*—factually incorrect generations produced with high confidence. Traditional detection methods relying on softmax probabilities (logits) or predictive entropy fail when models are miscalibrated ("confidently wrong"). This proposal introduces **Tangent-Space Consistency (TSC)**, a novel unsupervised detection framework grounded in the *Manifold Hypothesis*. We posit that truthful reasoning trajectories in the hidden state space $\mathcal{H}$ approximate a smooth, low-dimensional manifold, whereas hallucinations manifest as orthogonal departures from the local tangent space. By employing **Singular Value Decomposition (SVD)** to dynamically model the local subspace of the context window, TSC quantifies hallucinations as the Euclidean projection residual of the generated token. We propose to evaluate TSC against state-of-the-art baselines (e.g., SelfCheckGPT) on the **TruthfulQA** and **GSM8K** benchmarks, utilizing a quantized Llama-3-8B architecture on a constrained compute budget (NVIDIA P100).

1 Problem Statement & Motivation

Despite their capabilities, LLMs suffer from a critical reliability issue known as hallucination. Recent studies (Azaria & Mitchell, 2023) suggest that LLMs encode "truthfulness" in their internal representations, yet the final output layer (logits) often masks this signal due to Reinforcement Learning from Human Feedback (RLHF), which encourages confident stylistic output regardless of factual accuracy.

Existing detection methods have significant limitations:

1. **Logit-Based Methods (Perplexity):** Fail when the model assigns high probability to false tokens (calibration error).
2. **Sampling-Based Methods (e.g., SelfCheckGPT):** Require generating multiple samples ($N > 5$) per query, resulting in prohibitively high latency (Manakul et al., 2023).
3. **Supervised Probes:** Require training classifiers on labeled "truth/lie" datasets, which do not generalize well to new domains (Zou et al., 2023).

2 Theoretical Framework: The Geometry of Truth

2.1 The Manifold Hypothesis

We leverage the **Manifold Hypothesis** (Fefferman et al., 2016), which states that high-dimensional data (such as valid logical reasoning chains) concentrate near a lower-dimensional manifold $\mathcal{M}$ embedded within the ambient space $\mathbb{R}^D$.

- **Hypothesis 1:** Coherent, truthful generation follows a trajectory along the tangent bundle $T\mathcal{M}$.
- **Hypothesis 2:** Hallucinations represent a stochastic departure from $\mathcal{M}$ into the orthogonal complement $T\mathcal{M}^{\perp}$.

3 Methodology: Tangent-Space Consistency (TSC)

We propose an inference-time, unsupervised algorithm to detect these departures using linear algebraic projections.

3.1 Algorithm Definition

Let $h_t \in \mathbb{R}^D$ denote the hidden state vector of the LLM at time step t, where $D = 4096$ (for Llama-3-8B).

Step 1: Local Subspace Estimation

We define a context window W consisting of the preceding k tokens (e.g., $k = 10$):

$$H_{context} = [h_{t-k}, h_{t-k+1}, \ldots, h_{t-1}] \in \mathbb{R}^{k \times D} \tag{1}$$

We compute the **Singular Value Decomposition (SVD)** of the centered matrix $\bar{H}$:

$$\bar{H} = U \Sigma V^T \tag{2}$$

The top m right-singular vectors $V_m \in \mathbb{R}^{m \times D}$ span the *Local Tangent Space* of the reasoning trajectory.

Step 2: Projection & Residual Calculation

For the newly generated token h_t, we project it onto the subspace V_m. The projection operator P is given by $V_m^T V_m$. The residual vector r_t represents the component of the token *orthogonal* to the reasoning path:

$$r_t = h_t - (h_t \cdot V_m^T) V_m \tag{3}$$

Step 3: The TSC Score

The anomaly score is defined as the L_2 norm of the residual:

$$\mathcal{S}_{TSC}(t) = ||r_t||_2 \tag{4}$$

A spike in $\mathcal{S}_{TSC}$ indicates a **Geometric Hallucination**.

4 Novelty Analysis

Feature	**SOTA (SelfCheckGPT)**	**Proposed Method (TSC)**
Mechanism	Stochastic Sampling	Geometric Projection
Compute Cost	High (5× to 10× inference)	Negligible (One SVD per step)
Supervision	Unsupervised	Unsupervised
Basis	Uncertainty Quantification	Manifold Hypothesis

Table 1: Comparative analysis showing TSC's advantage in computational efficiency and theoretical grounding.

5 Experimental Design & Feasibility

5.1 Hardware Constraints

The experiments will be conducted on a **Single NVIDIA Tesla P100 (16GB VRAM)**. To accommodate the Llama-3-8B model within this budget, we will utilize **4-bit Quantization (NF4)** via the `bitsandbytes` library, reducing memory footprint from ∼16GB to ∼6GB, leaving sufficient VRAM for activation caching and SVD operations.

5.2 Datasets

- **TruthfulQA (Generation Task):** Specifically designed to elicit "imitative falsehoods." We will measure the AUROC of TSC in distinguishing correct answers from hallucinations.
- **GSM8K (Reasoning Task):** We will analyze if reasoning errors in multi-step math problems correlate with geometric departures in the hidden space.

6 References

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